Exam II Answers
Math 126 F Autumn 2018

1. (a) (4 points) $\dfrac{\partial z}{\partial x} = \dfrac{2y - 8x}{y^2}$ and $\dfrac{\partial z}{\partial y} = \dfrac{2x - 2yz + 3y^2}{y^2}$

   (b) (4 points) $z = -(x - 3) + \frac{3}{8}(y - 4) + 6$

2. absolute max $= 2$ (occurs at $(0, 1)$) and absolute min $= -\frac{1}{54}$ (occurs at $\left(\frac{1}{3}, \frac{1}{6}\right)$)

3. $V = \displaystyle\int_0^2 \int_{y/2}^{y} 8x + 2y \, dx \, dy$

4. $m = \displaystyle\int_{\pi/8}^{5\pi/8} \int_0^{\sin(2\theta) - \cos(2\theta)} 3r^2 \, dr \, d\theta$

5. $\displaystyle\int_0^9 \int_{\sqrt{x}}^{3} \cos(\pi y^3) \, dy \, dx = \int_0^3 \int_0^{y^2} \cos(\pi y^3) \, dx \, dy = 0$

6. $\displaystyle\iint_D 6y \, dA = \int_0^{\pi/3} \int_2^{4\cos\theta} 6r^2 \sin\theta \, dr \, d\theta = 22$